BOGYEOM PARK

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RESEARCH INTEREST

I design and study **AI-supported learning systems that promote deeper cognitive engagement**, at the intersection of **AI in Education, Human-AI Interaction, and Learning Analytics**.

- **Eliciting deeper engagement -** designing AI tutors that prompt learners to explain, compare, reflect, and build on dialogue rather than receive answers passively
- **Understanding learning processes -** analyzing conversational and behavioral traces to characterize cognitive engagement and connect interaction patterns with learning outcomes

EDUCATION

Seoul National University of Science and Technology — Seoul, South Korea
Integrated Ph.D. in Applied Artificial Intelligence — *Mar. 2023 - Present*
- Advisor: **Kyoungwon Seo**

Seoul National University of Science and Technology — Seoul, South Korea
B.S. in Electrical and Information Engineering — *Mar. 2019 - Feb. 2023*
- GPA: 4.14/4.5 (Major GPA: 4.5/4.5)
- Thesis: Online Learning Support System Based on Facial Recognition

HONORS & AWARDS

- **AI SeoulTech Graduate Scholarship** (KRW 10,000,000), Seoul Scholarship Foundation, 2025
- **Best Student Paper Award**, IEEE Seoul Section, 2024
- **National Science and Engineering Undergraduate Scholarship** (full tuition), Korea Student Aid Foundation, 2021-2022
- **Best Capstone Design Award**, Seoul National University of Science and Technology, 2022

REFEREED JOURNAL ARTICLE

Integrating Biomarkers From Virtual Reality and Magnetic Resonance Imaging for the Early Detection of Mild Cognitive Impairment Using a Multimodal Learning Approach: Validation Study
Bogyeom Park, Yujin Kim, Jiwoo Park, Hyunbin Choi, Se Eun Kim, Hojin Ryu, and Kyoungwon Seo
Journal of Medical Internet Research, 26, e54538 (2024) **SCIE; JCR Top 3%; Q1**

EXTENDED ABSTRACTS

Assessing Critical Thinking Through a Multi-Agent LLM-Based Debate Chatbot
Bogyeom Park and Kyoungwon Seo
CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems

Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment
Bogyeom Park, Yujin Kim, Jiwoo Park, Hyunbin Choi, Se Eun Kim, Hojin Ryu, and Kyoungwon Seo
CHI EA '24: Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems

RESEARCH EXPERIENCE

Human-centered Artificial Intelligence (HAI) Lab, SeoulTech — Seoul, South Korea
Graduate Researcher (Advisor: Kyoungwon Seo) — *Mar. 2023 - Present*

ICAP-Based AI Tutoring System for Probability and Statistics Learning
Lead Researcher | Korea Education & Research Information Service — *Mar. 2026 - Aug. 2026*
- Designed and evaluated an ICAP-based AI tutor that uses staged elicitation to promote active, constructive, and interactive engagement rather than answer delivery
- Developed an utterance-level coding and analytics workflow linking dialogue evidence with tutor correctness, usage patterns, and learning outcomes
- Built an interactive research dashboard for reviewing engagement labels and comparing AI-tutored learning processes

GUI Agent Technologies for Automated UX Accessibility Evaluation
Research Assistant | National Research Foundation of Korea *Sep. 2025 - Present*
- Designed a GUI agent capable of performing expert-level automated evaluations of UX accessibility
- Planned experimental protocols for validating human-AI comparative performance in accessibility assessments

Agentic AI for Personalized Career, Academic, and Counseling Support
Lead Researcher | Korean Educational Development Institute *Jun. 2025 - Dec. 2025*
- Investigated how agentic AI could support personalized educational pathways, counseling, and decision-making

AI Copilot Technologies for Adaptive, Teacher-Augmented Learning
Research Assistant | Institute for Information & Communications Technology Planning & Evaluation *Jul. 2023 - Aug. 2025*
- Built counseling chatbot and analysis models supporting AI-driven student coaching
- Co-developed an integrated platform enabling personalized teacher-augmented learning

Human-centered Artificial Intelligence (HAI) Lab, SeoulTech Seoul, South Korea
Undergraduate Researcher (Advisor: Kyoungwon Seo) *Jul. 2021 - Feb. 2023*

LLMs to Support Teachers in Educational Settings
Research Assistant | Lab Project *Mar. 2023 - Feb. 2024*
- Fine-tuned an LLM to generate student competency-analysis reports and assessed its usefulness through expert interviews
- Identified opportunities and limitations of LLM support for competency assessment and report generation

Multimodal Digital Biomarkers for Early Dementia Diagnosis
Research Assistant | National Research Foundation of Korea *Mar. 2022 - Feb. 2024*
- Collected VR kiosk interaction data from participants with mild cognitive impairment and healthy controls in collaboration with Hanyang University Guri Hospital
- Developed multimodal predictive models integrating VR behavioral and MRI biomarkers for early detection of mild cognitive impairment

TEACHING

Deep Learning Fall 2023
Teaching Assistant, Seoul National University of Science and Technology
- Designed a final project using CNN-based models to predict drivers' physical and cognitive states from image data
- Supported lectures, advised student projects, and graded assignments and examinations

SKILLS

- **Research Methods -** experimental design, usability evaluation, user research, prototyping, multimodal learning analytics, statistical analysis, qualitative coding
- **AI and Data -** machine learning, deep learning, feature engineering, LLM applications, AI agents
- **Programming and Analysis -** Python, C/C#, SPSS
- **Design -** HCI theory, UX design, interaction prototyping, usability testing

PATENT

Explainable AI-Based System for Early Diagnosis and Prognosis Prediction of Alzheimer's Disease Using VR Biomarkers, and Method Thereof (Korean Patent Application No. 10-2023-0105821, filed Aug. 2023)

SERVICE

- **Publicity Manager**, SeoulTech Human-centered Artificial Intelligence Lab